	Academic Semester	Fall 2026-2027	**Academic level**	3
	Course Code	MT301	**Exam**	Quiz 1 — Model A
	Course Title	Probability and Statistics I		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Mai Farag , Dr. Walid Abd-Elraheem		

Student Name: _____ **Student ID:** _____

*Answer **all** questions. Calculators are allowed. Show your work in Questions 2 and 3; give probabilities as simplified fractions.*

Question 1 — Choose the correct answer (4 points)

One point each: ½ for the correct letter, ½ for a correct reason. Circle the letter and write why you chose it in the box.

1. A restaurant offers 4 appetizers, 6 main dishes and 3 desserts. How many different meals consisting of one appetizer, one main dish and one dessert can be ordered?

A) 13 **B)** 24 **C)** 36 **D)** 72

Why? (reason or calculation)

2. A club of 10 members elects a president, a vice-president and a secretary (one person per post). In how many ways can this be done?

A) 720 **B)** 120 **C)** 1000 **D)** 30

Why? (reason or calculation)

3. Two fair dice are rolled. The probability that the sum of the two faces is 8 is

A) $\frac{1}{6}$ **B)** $\frac{1}{9}$ **C)** $\frac{5}{36}$ **D)** $\frac{7}{36}$

Why? (reason or calculation)

4. A fair coin is tossed three times. The probability of getting **exactly** two heads is

A) $\frac{3}{8}$ **B)** $\frac{1}{4}$ **C)** $\frac{1}{2}$ **D)** $\frac{2}{3}$

Why? (reason or calculation)

Question 2 — Counting (3 points)

A committee of 4 people is to be chosen from 6 men and 5 women.

(a) In how many ways can the committee be chosen? [1.5 pt]

(b) In how many ways can it be chosen if it must contain **exactly 2 women**? [1.5 pt]

Question 3 — Probability (3 points)

A 3-digit number is formed at random from the digits $\{1, 2, 3, 4, 5, 6\}$, with **no digit repeated**.

(a) Find the probability that the number is **divisible by 5**. [1.5 pt]

(b) Find the probability that the number is **greater than 400**. [1.5 pt]

— End of quiz · Good luck —